

Web Technologies – FEUP-MM0047-2025/2026

Project Installation Guidelines

GEO-SHOOTER: Battle Arena

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January 2026

1 Overview

GEO-SHOOTER is a multiplayer battle arena game designed for **local competitive play**. Players use their **mobile devices as motion-sensitive controllers** (gyroscope/accelerometer) to navigate and battle on a central PC screen.

1.1 Key Features

- **Multiplayer Arena:** battle against friends in real-time using a room-based matchmaking system.
- **Motion Control:** smartphone gyroscope and accelerometer drive character movement.
- **Multiple Character Classes:** five distinct fighters (Tank, Sniper, Runner, Balanced, Assault) with unique stats and special abilities.
- **Dynamic Maps:** six arenas with hazards such as toxic zones, moving walls, and gravity anomalies.
- **Immersive Audio:** background music and SFX for gunfire, power-ups, and respawns.

1.2 Tech Stack

- **Backend:** Node.js + Express
- **Communication:** WebSockets (`ws`) for low-latency input and game state synchronization
- **Frontend:** HTML5 Canvas, Vanilla JavaScript, CSS3
- **Libraries:**
 - `Chart.js`: controller UI visualizations (e.g., stats charts)
 - `qrcode.min.js`: QR code generation for fast mobile join

2 Project Structure

Ensure the files are organized as follows (root directory contains `server.js`):

```
/
certs/ # Required for HTTPS (motion sensor support)
  cert.pem
  key.pem
public/ # Static assets served by Express
fonts/
```

```
scripts/ # Client-side JS (landing.js, game.js, controller.js)
sounds/ # Audio files (.mp3)
styles/ # CSS files (game.css, landing.css, controller.css)
vendor/ # Third-party scripts (e.g., qrcode.min.js)
landing.html # Main entry point
game.html # Main game arena
controller.html # Mobile controller interface
server.js # Node.js backend server
package.json # Project dependencies
```

3 Installation Guide

3.1 1. Get the Project Files

You can run the project either by downloading a ZIP or cloning a repository.

Option A – ZIP:

- Download the project ZIP to your machine
- Extract it to a folder (this will be the project root)

Option B – Git clone (if provided):

```
git clone <REPOSITORY-URL>
cd GEO-SHOOTER
```

3.2 2. Prerequisites

Make sure you have:

- **Node.js** (v14 or higher)
- **npm** (bundled with Node.js)

Check versions:

```
node -v
npm -v
```

3.3 3. Install Dependencies

From the project root:

```
npm install
```

Note: If your `package.json` does not include dependencies yet, you can initialize and install:

```
npm init -y
npm install express ws
```

3.4 4. Setup SSL (Crucial for Mobile Motion Sensors)

Why HTTPS is required: mobile browsers (especially iOS Safari) block gyroscope/accelerometer access on insecure origins. For this reason, **HTTPS is mandatory**.

3.4.1 4.1 Create the certs/ folder

From the project root:

```
mkdir -p certs
```

3.4.2 4.2 Find your local IP address

You must use your PC IP in the same Wi-Fi network to let phones connect.

macOS (Wi-Fi):

```
ipconfig getifaddr en0
```

Linux:

```
hostname -I
```

Example output: 192.168.1.25

3.4.3 4.3 Method A (simple): self-signed cert (works for local LAN)

Replace YOUR-IP with your local IP (e.g., 192.168.1.25):

```
openssl req -x509 -newkey rsa:2048 -nodes \  
-keyout certs/key.pem \  
-out certs/cert.pem \  
-days 365 \  
-subj "/C=PT/ST=Porto/L=Porto/O=FEUP/OU=WebTech/CN=YOUR-IP"
```

3.4.4 4.4 Method B (recommended): self-signed with SAN

Browsers often expect **Subject Alternative Name (SAN)**. This method improves compatibility.

Create config file:

```
cat > certs/openssl-san.cnf <<'EOF'  
[req]  
default_bits = 2048  
prompt = no  
default_md = sha256  
distinguished_name = dn  
x509_extensions = v3_req  
  
[dn]  
C = PT  
ST = Porto  
L = Porto  
O = FEUP  
OU = WebTech  
CN = GEO-SHOOTER  
  
[v3_req]  
subjectAltName = @alt_names  
  
[alt_names]  
DNS.1 = localhost  
IP.1 = 127.0.0.1  
IP.2 = YOUR-IP  
EOF
```

Edit YOUR-IP to your real IP, for example using:

```
sed -i.bak "s/YOUR-IP/192.168.1.25/g" certs/openssl-san.cnf
```

Generate cert + key:

```
openssl req -x509 -newkey rsa:2048 -nodes \  
-keyout certs/key.pem \  
-out certs/cert.pem \  
-days 365 \  
-config certs/openssl-san.cnf
```

3.5 5. Run the Server

Start the application from the project root:

```
node server.js
```

The server runs on:

- <https://localhost:8443> (same machine)
- <https://<YOUR-IP>:8443> (other devices on the same network)

3.6 6. Network Requirements (Important)

- PC and phones must be on the **same Wi-Fi network**
- Allow incoming connections on port **8443** (Firewall may block)

If phones cannot connect, check:

- Your IP is correct and stable (some networks change it)
- Firewall rules allow inbound traffic on 8443

4 How to Play

4.1 1. Launch the Game (PC)

On the computer, open:

- <https://<YOUR-IP>:8443> (recommended for mobile connectivity)
- or <https://localhost:8443> (PC-only testing)

4.2 2. Access the Controller (Mobile)

On the phone, either:

- Scan the QR code displayed on the screen or open directly: <https://<YOUR-IP>:8443/phone>

4.3 3. Join a Room

Enter the same **Room ID** on both the PC and the phone.

4.4 4. Select Character

Choose one fighter on the phone:

- **Tank:** high health, low speed, *Shield* ability
- **Sniper:** high damage, long range, *Piercing Shot* ability
- **Runner:** extreme speed, low health, *Dash* ability
- **Balanced:** all-around stats, *Heal Burst* ability
- **Assault:** high fire rate, *Bullet Storm* ability

4.5 5. Calibrate & Battle

- Follow the on-screen prompts to calibrate phone sensors
- Tilt the phone to move
- Tap **FIRE** or **ABILITY** to play

5 Troubleshooting: iOS & Motion Sensors

5.1 1. HTTPS is mandatory

- Motion sensors are blocked on HTTP
- Always use: `https://<YOUR-IP>:8443`

5.2 2. Accept the self-signed certificate

On iPhone Safari:

- When the warning appears, open advanced options and continue
- If motion still does not work, ensure the certificate is trusted (if available):
 - Settings → General → About → Certificate Trust Settings

5.3 3. Enable Motion & Orientation Access

- Check Safari settings and allow **Motion & Orientation Access** if present

6 Notes

- If multiple devices fail to connect, try restarting the server and reloading pages
- For best results, keep the phone screen awake during gameplay
- If your Wi-Fi isolates clients (some public networks do), phones may not reach the PC